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Implementation of an Online Surface Network Model (SNM) for Real-Time Production Optimization

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Abstract

Objectives/Scope: This paper presents an implementation of an Online Surface Network Model (SNM) designed to provide real-time visibility and optimization capabilities across a brownfield surface production system. The objective of the trial was to evaluate the feasibility, technical readiness, and value potential of integrating live field data with a hydraulic surface network model. The initiative aimed to enhance operational decision-making, identify flow bottlenecks, and support production optimization through a digital twin of the surface network.

Methods, Procedures, Process: The trial involved modeling a representative section of the surface network—including wells, flowlines, manifolds, etc.—using a hydraulic simulation modelling tool integrated with real time data and updated digital twins of associated wells and PVT's. Calibration of the model was carried out using historical and real-time data to ensure accuracy in pressure and flow predictions. A collaborative approach was adopted, involving production operations, subsurface teams, and digital specialists. The project also focused on aligning the SNM output with field measurements and identifying potential operational scenarios that could be simulated to support decision-making.

Results, Observations, Conclusions: The implementation of the Online Surface Network Model (SNM) demonstrated significant value in enhancing both the speed and quality of operational decision-making. Traditionally, engineering teams relied on offline models, which often required several days to compile data, perform simulations, and communicate results. With the SNM integrated directly into the live data environment, the team was able to perform simulations and analyze network scenarios in near real-time, greatly improving responsiveness and reducing turnaround time from days to minutes. A major advantage observed during the project was the model's ability to visualize and quantify well-level and node-level performance, allowing for granular tracking of production inefficiencies and bottlenecks. Using this insight, the team was able to identify and prioritize a set of optimization actions with the potential to deliver oil gains equivalent to approximately 2% of total station production—a considerable opportunity, especially within a brownfield context. Beyond production improvements, the model also enhanced well-test planning, supported proactive maintenance scheduling, and improved coordination between field

operations, production engineering, and planning teams. These results reinforced the SNM's role not just as a simulation tool, but as a practical, decision-support solution with the potential to scale across larger network areas and contribute to long-term value realization.

Novel/Additive Information: This project represents an early-stage application of a live-integrated SNM within a brownfield environment. Unlike traditional static models, it provided real-time operational insights and enabled virtual metering for non-instrumented sections. The project also explored integration with production allocation systems, offering a practical pathway for future scalability. The experience highlights both the technical and organizational considerations essential for adopting digital twin technologies in legacy surface networks.

Introduction

Improving efficiency in the energy sector, particularly brownfield surface production networks, is a significant issue. Assets are aging, and infrastructure is becoming increasingly complex, making it essential to monitor and optimize the performance of surface networks to maximize production and lower operating costs. Traditional methods of network modeling and optimization are static and offline-driven, which are time-consuming and not aligned with real-time field environments. These constraints typically result in slow decision-making that does not enable operators to respond in real time to changes in production dynamics.

The developments in digital technologies, particularly the integration of real-time data with advanced simulation tools, offer a groundbreaking fix to this issue. The use of digital twins, for instance, allows for the creation of live, virtual representations of physical systems that can be continuously updated with real-time data, providing operators with immediate and actionable insights. This paper introduces the implementation of an Online Surface Network Model (SNM) that leverages these technologies to offer real-time visibility and optimization capabilities for brownfield production systems. Through the duplication of the surface network—wells, flowlines, and manifolds (MSV's)—this technology enables a greater in-depth view of the system's performance, highlighting potential bottlenecks and inefficiencies that would otherwise go unnoticed.

The primary purpose of this trial was to determine the technical feasibility and preparedness to integrate hydraulic surface network models and real-time field data to enhance decision-making activities, identify flow limitations, and enable further production optimization activities. By reducing the need for hand simulations and providing instant feedback, this integration enables improved quality and velocity of decision-making. Before the trial, the authors conducted an exhaustive review of advanced engineering literature and industrial case studies of real-time network modeling, digital twin technology, and brownfield optimization methods. The most important findings elicited in this literature were systematically synthesized and adapted to suit the specific problems and constraints of the field environment studied in this study. An overview of this literature review, detailing how it relates to the design of the model and its implementation plan, is presented in the following Literature review section. The approach also has the capability to justify maintenance scheduling, well-test planning, and inter-team coordination.

The paper outlines the trial methodology, such as model calibration with historical and real-time data, and a collaborative framework implemented across production operations, subsurface groups, and digital experts. The paper also touches upon outcomes from the trial, which pointed out some of the major advantages of the integrated approach, such as turnaround times for simulations being lower, improved identification of production inefficiencies, and improved coordination between the involved teams.

This paper is a preliminary trailblazing step towards the application of real-time SNM's within brownfield settings and presents significant technical and organizational considerations necessary for integrating digital twin technology with legacy surface networks.

Literature Survey

The integration of real-time information in the operation of production has been a topic of strong interest over the past few decades, particularly with growing importance of digital transformation in the oil and gas industry. There have been some key studies and industry reports that talked about the prospects of using integrated models for surface network optimization, enhancing decision-making, and operational simplification.

Early work in the field of oilfield optimization by real-time data integration is attributed to [McKie et al. \(2001\)](#), wherein they discussed the application of mathematical techniques for production system optimization. They emphasized how the application of integrated models could reduce complex production systems and reduce inefficiencies. This was followed by the work of [Nordtvedt and Unneland \(2006\)](#), who covered real-time asset management from a practical perspective, offering insights into how real-time information and automation had the potential to improve decision-making and reduce operational costs in asset management systems.

The development of integrated field planning tools was also emphasized, especially in multi-field projects. [Schneider et al. \(2002\)](#) applied integrated field planning to a multi-field gas development in offshore Southeast Sumatra. This study demonstrated the power of unified models for managing complex, multi-dimensional production schemes, providing an architecture that could be evolved into real-time optimization workflows.

A further step towards more integrated production systems was seen in the work of [Rosales et al. \(2006\)](#), who presented the implementation of subsurface-to-surface workflows for field exploitation strategies in the Litoral De Tabasco Asset. The integration of real-time data and production models was seen to improve the technical and economic performance of production systems, providing the foundation for further development in the use of integrated workflows. Similarly, [Cetkovic \(2007\)](#) emphasized the need for treating facilities, wells, and reservoirs as a system, and advocated for holistic modeling approaches that reflect real-time dynamics across production assets.

The concept of online production optimization began to emerge more clearly in the mid-2000s, with [Shere et al. \(2008\)](#) reporting online optimization techniques at the Ekofisk field. The paper illustrated that integrating real-time data and optimization models had the potential to greatly increase production throughput, reduce downtime, and facilitate faster decision-making in response to changing field conditions.

At the same time, the overall trend towards digital oilfield (DOF) implementations was explored in a number of studies. For example, [Piovesan and Kozman \(2011\)](#) emphasized artificial intelligence innovations and their application within the oilfield context. Their research indicated that AI-based models, when combined with real-time data, had the potential to revolutionize the decision-making process by providing predictive analysis of production performance. Likewise, the research by [Ershaghi and Al-Abbassi \(2012\)](#) emphasized the shift from conventional management hierarchies to digital cultures among national oil companies. This was at the center of the adoption of integrated production models and full realization of digital oilfield capabilities.

As real-time data and model integration technology matured, the benefits were seen ever more in large-scale applications. In particular, [Desai et al. \(2016\)](#) examined the successful integration of digital field technology in the Greater Burgan Field, Kuwait. They presented work on how integrated technologies would make production optimization, reservoir management, and operational workflows easier. Additionally, [Reddicharla et al. \(2019\)](#) addressed data and model integration issues for production forecasting in super-giant fields. They highlighted the necessity of utilizing integrated models in handling large, mature fields' complexities while making reliable production forecasts.

The use of real-time, data-driven models was also boosted by the increased focus on production optimization and prediction. [Reddicharla et al. \(2020\)](#) provided a complete review of holistic data management and architecture philosophies, including the lessons learned from digital oilfield

implementations in giant brownfields. Their article emphasized how real-time integration can provide the overall asset performance view and enable proactive management decisions in large, multi-field environments. Further research by [Alsaeedi et al. \(2020\)](#) examined end-to-end integration platforms that linked reservoir and surface network data to predict short-term production. The approach demonstrated potential for utilizing real-time data to evaluate production deliverability and optimize production targets.

Since the interest was focused on streamlining production in a more scalable manner, [Al Jabri et al. \(2024\)](#) presented an automated production allocation and deferment calculation system founded on integrated digital oil field surveillance data and physics-based well models. The system improved the accuracy and timeliness of decision-making through the automation of difficult tasks such as allocation and deferment calculations, a significant move towards achieving real-time optimization at scale and capable to integrate with the surface networks.

The importance of integration and data management in optimizing production workflows is also underlined by studies like that of [Reddicharla et al. \(2020\)](#), who spoke about integrated data management architectures and best practices in giant brownfields. These writings collectively illustrate that integrated systems, as complemented by real-time data, yield significant advantages both in terms of efficiency and production optimization in mature field settings.

Objectives

The primary aim of this study is to explore the technical readiness, feasibility, and value potential of integrating real-time field data with a surface network model (SNM) for optimized production of oil in brownfield. The following crucial objectives were set to guide the study and ensure comprehensive evaluation of the online surface network model:

Assess Feasibility of Real-Time Data Integration into Surface Network Modeling

Part of the principal objectives of the current study was to find out the technical feasibility of feeding real-time operation data into a hydraulic simulation model of the surface network automatically. This involved an examination of the capability of the existing network infrastructure to carry out real-time data transmission and integration and ensure flow and pressure predictions accuracy and dependability. The review was about the flexibility of the system for brownfield environments with regard to, first and foremost, legacy infrastructures and data availability.

Assess the Impact of Digital Twin Technology on Production Optimization

It is one of the research goals where the capability of digital twin technology, e.g., the creation of a dynamic, live-represented digital twin of the surface network (working on a thin client application accessible by multiple users at same time), to aid in production optimization activities was evaluated. This consists of the identification of bottlenecks and inefficiencies in production and the network through simulation and integration of real-time information. The objective was to demonstrate how this digital twin can assist in operational decision-making, reduce downtime, and identify optimization opportunities that would not readily be identifiable using traditional offline modeling techniques.

Identify Opportunities for Operational Improvements and Optimization

This objective was to utilize the surface network model (SNM) on the thin client application for some production inefficiencies, bottleneck opportunities, and optimization potentials in the surface network. The objective was to quantify and rank potential optimization activities that could foreseeably deliver substantial production increases. The trial was designed to demonstrate that real-time simulation could provide insightful assistance in optimizing well production, planning maintenance schedules, and improving coordination among the production teams. Focus was placed upon the assessment of the feasibility of an increase in Facility-A oil output.

Explore the Scalability and Long-Term Value of SNM in Brownfield Settings

The final objective was to explore the scalability of the SNM for future deployment in larger network regions and the extent to which it can influence long-term value realization in mature, brownfield assets. This included challenging the model's ability to scale past the initial trial area and its role in proactive planning of maintenance, reservoir management, and optimizing field coordination in general. The ambition here was to figure out how technology in digital twins, once deployed, could become the basis for future decision-support systems within big, complicated production environments.

Methodology

The methodology for the trial was built around cutting-edge technologies and involved the integration of real-time data, sensors, and predictive algorithms to enhance operational capabilities. Key components of the methodology included:

Well Mapping

For the trial Facility-A, physics-driven digital twin well models were created for every well within the Well Management Visualization System (WMVS). Quality checking (QC) of the models was performed automatically by a workflow established by the Petroleum Engineering team prior to usage as Well-nodes within the SN model. In our case, since the well modeling process has been performed within an existing legacy process that has been operational since 2015, much time was conserved since existing models were ready to be used.

Real-time sensor data collection and integration with WMVS & digital twin models

Advanced sensors are deployed across critical components of the wells and surface infrastructure. These sensors measured key operational parameters such as:

- Well operating parameters (ESP, PCP, BP running parameters including pressure for surface and downhole nodes)
- Downstream Flow rate: Monitored to ensure optimal transportation of fluids at intermittent stations and custody points.
- Network Pressure: Measured to detect deviations and track safe operating conditions.

Data was transmitted in real time to a centralized data management platform which is integrated with legacy WMVS, where it was processed and analyzed to generate actionable insights for project team. (Figure 1)

Data present in PDO GIS maps and static datasets (.pdf files, old standalone network models, excel files, etc.) was studied and gathered in a structured digital database. Same data was discussed with Process optimization focal point for their confirmation on latest data and then used for Network mapping exercise. An initiative project is underway for all PDO assets to convert historic surface network data to an online database to enable automated, periodic updating for use in real-time surface network models.

Real-time ESP, PCP, BP, and other well-type sensor data for the trial Facility-A, along with validated well test data for all wells, is transmitted to the SCADA system and incorporated into the legacy WMVS. This data feeds into mapped physics-based digital twin well models, which are used to update well and pump performance curves daily, enabling continuous tuning of the models to real-time conditions for accurate history match & simulation within the online surface network model.

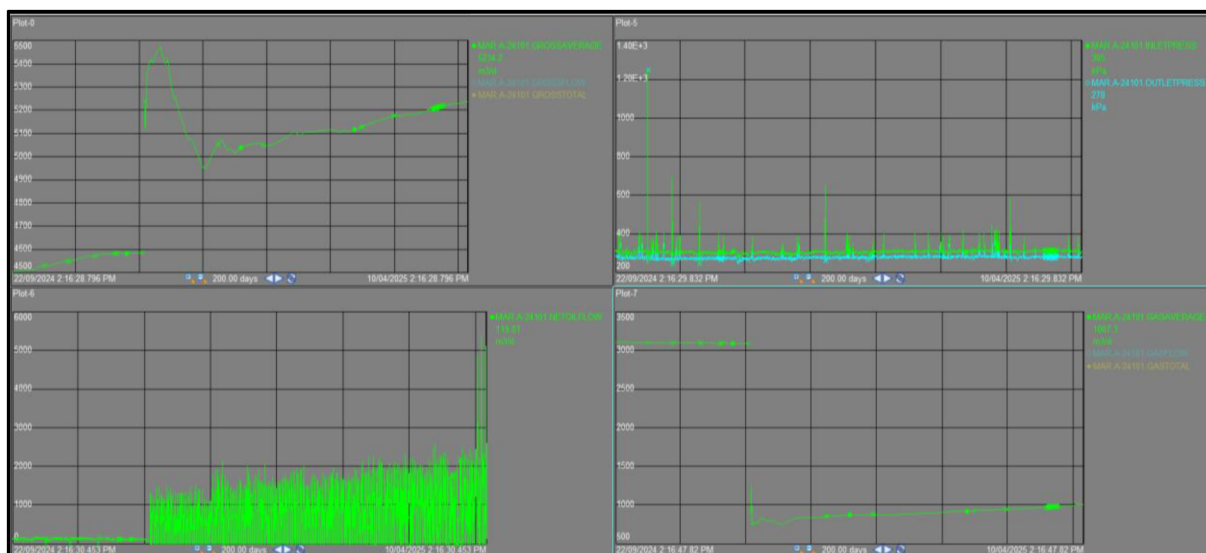


Figure 1—Example of Sensor data (Pressure and Rates) at critical surface network nodes

Surface Network Modelling (SNM) creation

A digital twin of the surface infrastructure was developed within a thin-client environment, enabling real-time creation, updates, and maintenance of the surface network model (Figure 2). This digital representation provided a visual model of the entire network, allowing engineers to understand the interdependencies between different components. The digital twin facilitated real-time monitoring of the network, which allowed for more effective decision-making and better identification of inefficiencies or risks.

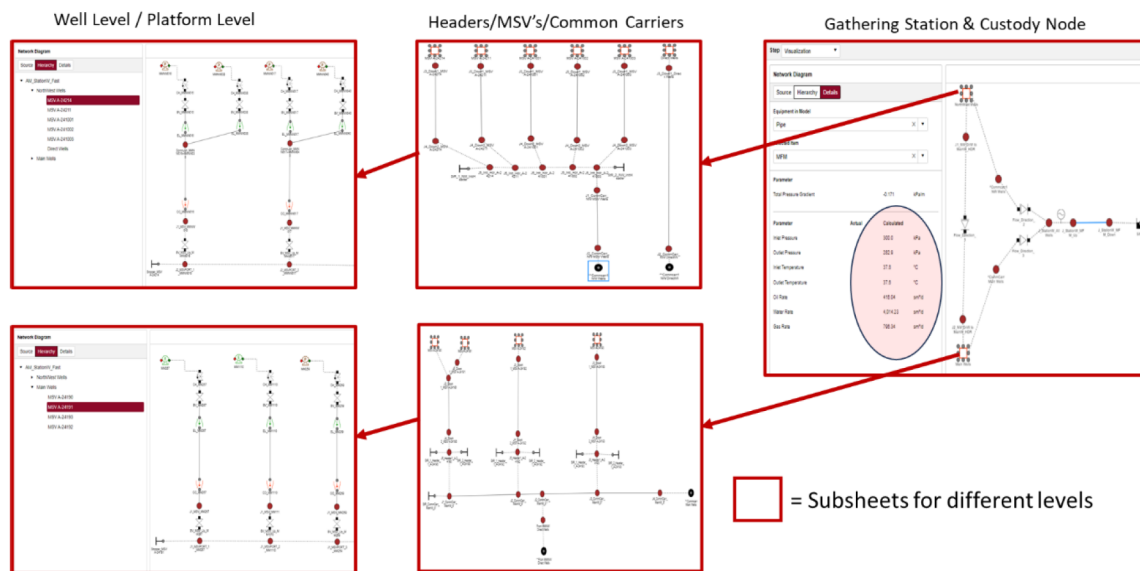


Figure 2—Created – Surface Network Model

As illustrated in Figure 2, the surface network model (SNM) was developed in three main sections:

- Well Pad Level (Figure 3) – modeling was carried out from individual well nodes to the outlet of the well pads, capturing all pad-level equipment.
- MSV/Header Level (Figure 4) – this section encompassed the SNM from the well pad outlets to the MSVs/headers, incorporating varying pipeline diameters, elevation profiles, and intermittent flow stations.

- c. Gathering Station & Custody Node Level (Figure 5) – this included modeling from the headers to the custody transfer point, with detailed representation of separators and other gathering station infrastructure.

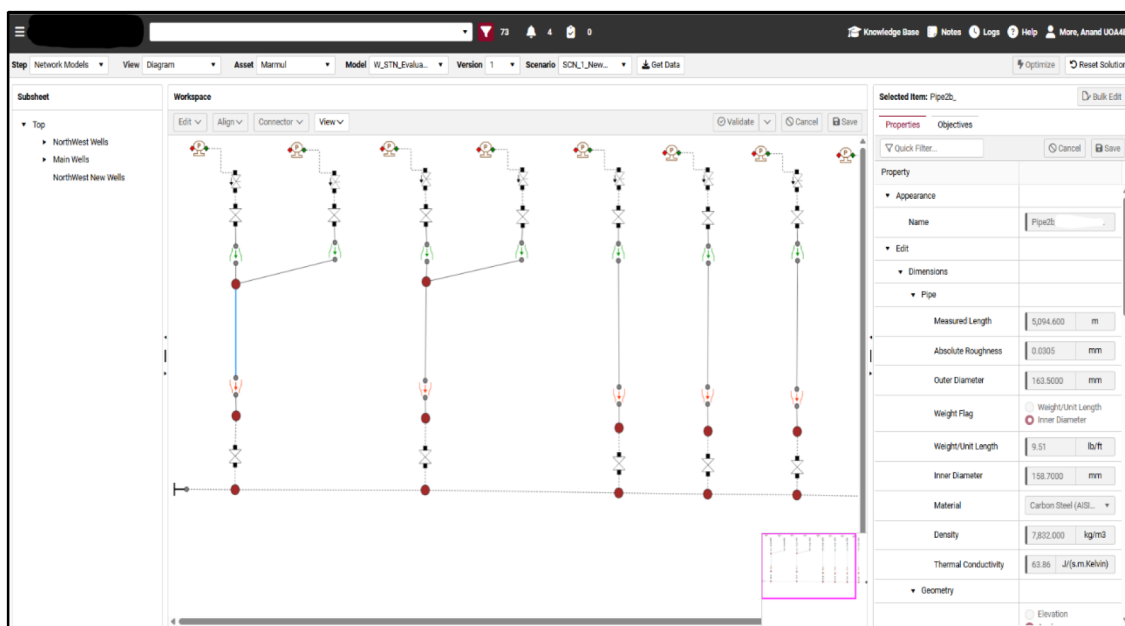


Figure 3—Well Pad Level SNM

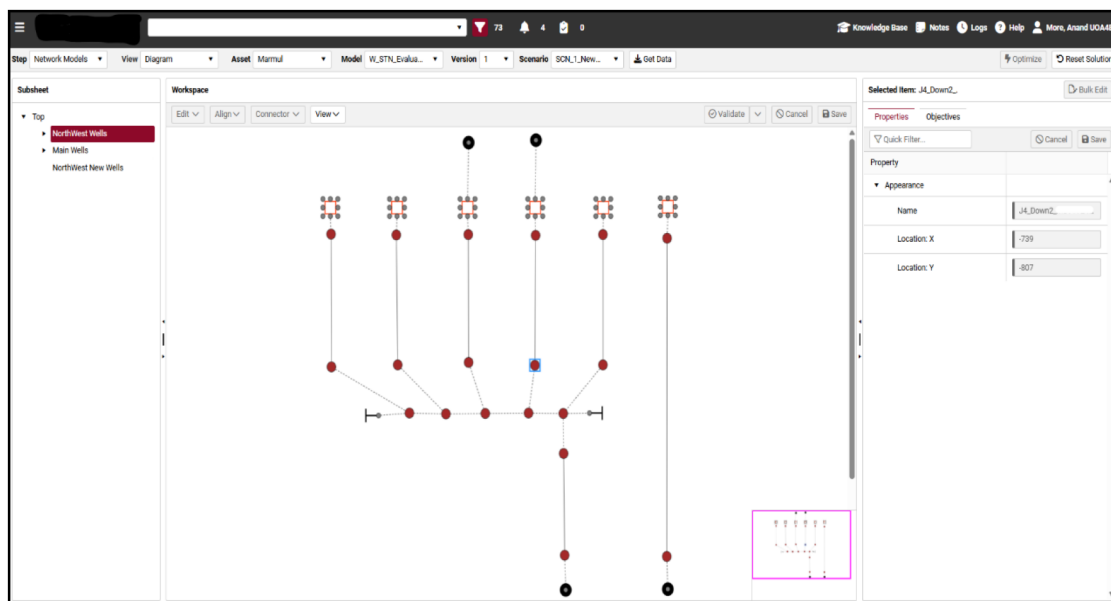


Figure 4—MSV/Header Level SNM

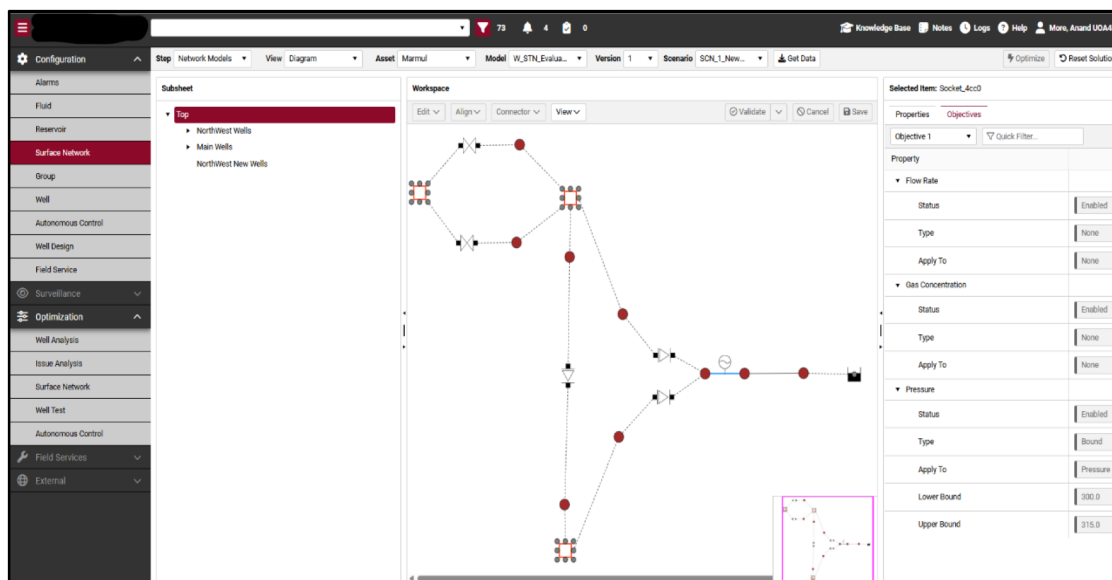


Figure 5—Gathering Station & Custody Node Level SNM

History Matching of the created SNM

As outlined earlier in this paper, real-time data, along with the most recent valid well test data, were utilized to update the physics-based digital twin well models. These well models, integrated into the created SN model, were converged to establish the base scenario for conducting the history matching exercise, thereby ensuring confidence in the accuracy of the created surface network model (SNM). Below Figure 6 shows the accepted results of the SNM base scenario history match.

Parameter	Actual	SNM	Match %
Rate (Sm ³ /d)	Gross: 4510	Gross: 4430 Oil: 416 Water: 4014	98%
MFM Pressure (kpa)	315	300	96%

Figure 6—SNM – History match Results

Post acceptance of the history match results and well hook-up details with all the stakeholders, custom optimization scenario's: a) New Wells case and b) Oil Maximization case was prepared to work on main objective 'Identify Opportunities for Operational Improvements and Optimization (Oil Gain)' of the trial.

Automation of daily SNM convergence (Daily match scenario and custom optimization scenarios)

We automated the daily convergence of SNM, including the daily match and custom optimization scenarios. Apart from ensuring the model was in sync with real-time data, this facilitated ease of timely delivery of daily results. The automation improved convergence rate, decision making, and general reservoir management activities.

Real-Time Monitoring and Daily Evaluation Dashboard creations

We developed a Real-Time Monitoring and Daily Evaluation Dashboard (Figure 7) to track daily results and real-time data updates, streamlining model performance insights and decision-making for optimized reservoir operations.

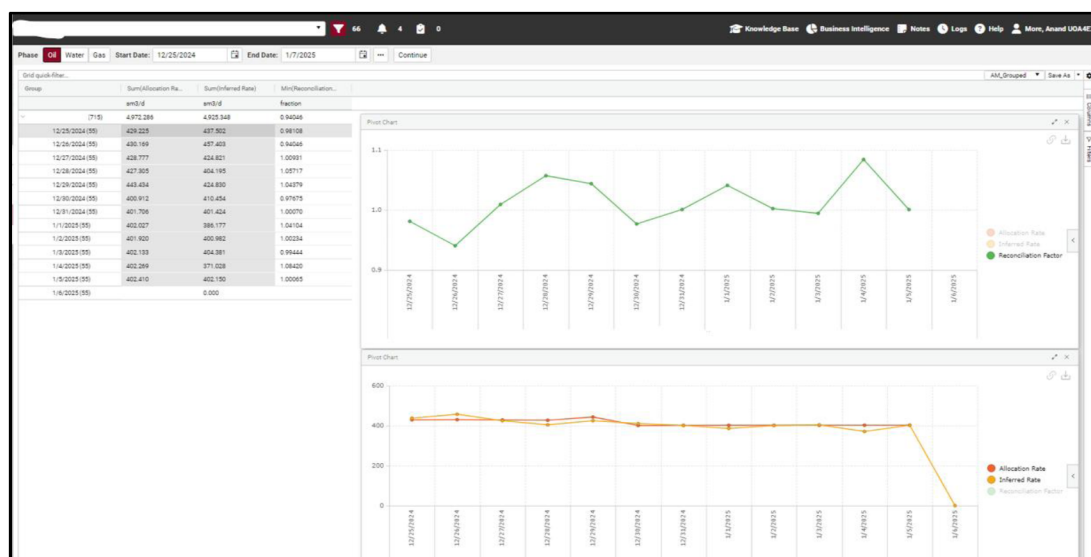


Figure 7—Real-Time Monitoring and Daily Evaluation Dashboard

Results and Discussion

The development and deployment of the Online Surface Network Model (SNM) have provided valuable insights into its operational effectiveness and collaborative potential. The model, executed in a thin client application, operates autonomously on a daily basis without constant human intervention. This common platform facilitates multiple users to access and contribute to the system concurrently, making real-time data sharing and decision-making more effective. One of the primary advantages of its application has been a 2.97% oil gain for the Facility-A under trial during the evaluation phase, demonstrating the immediate impact and potential of SNM in optimizing reservoir management. The outcome will be discussed in detail in this section, coupled with the implications of future integration of Machine Learning (ML) and Artificial Intelligence (AI), which will further enhance the functionality of SNM in subsequent phases.

Actions on Wells basis SNM optimization scenario output

Well Optimization: Actions were recommended for 4 wells out of 53 producers in the trial Facility-A, to reduce back pressure through choke optimization at well level. This scenario was run as a custom conditional oil maximization scenario.

- For Well A and Well B (Figure 8): Actions taken resulted in oil gains of 0.87 m³ and 8.89 m³, respectively. These gains were validated by performing a mobile separator (PI test) on these two wells. The total oil gain accounts for 2.67% of the Net Oil gain of Facility-A total oil.
- For Well C and Well D: The choke optimization helped in pressure maintenance of the network downstream to the well chokes improving production from the commingled wells and these wells itself, improving net oil. The total oil gain due to this accounts for 0.3% of the Net Oil gain of Facility-A total oil.

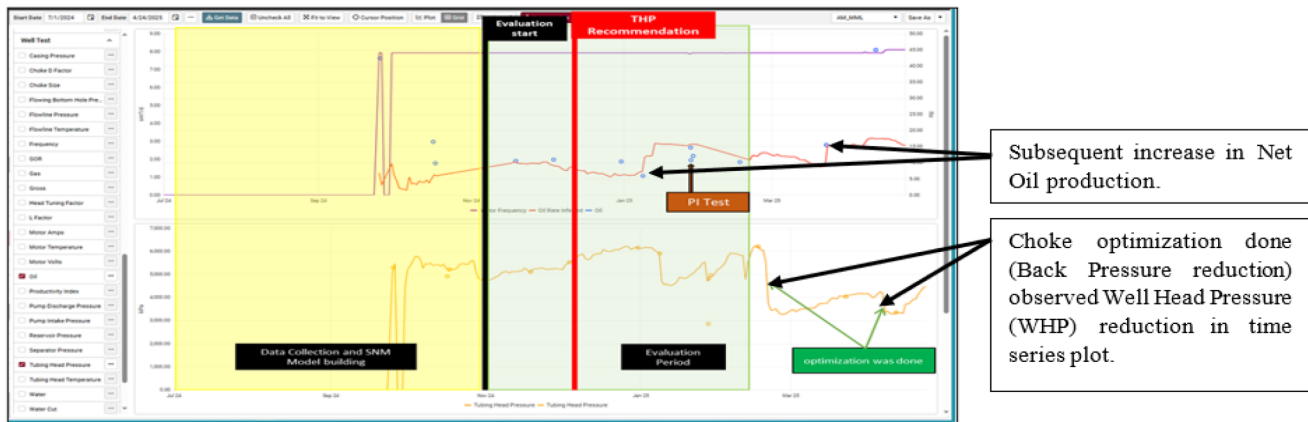


Figure 8—Well B Optimization Action and Oil Gain (Time series plot of Well B) – During Trial Evaluation period

Asset level – SN Model Oil production results and Oil Gains for trial Facility-A

Figure 9 shows the results of SNM optimization over the entire evaluation period, demonstrating sustained oil gains throughout the duration.

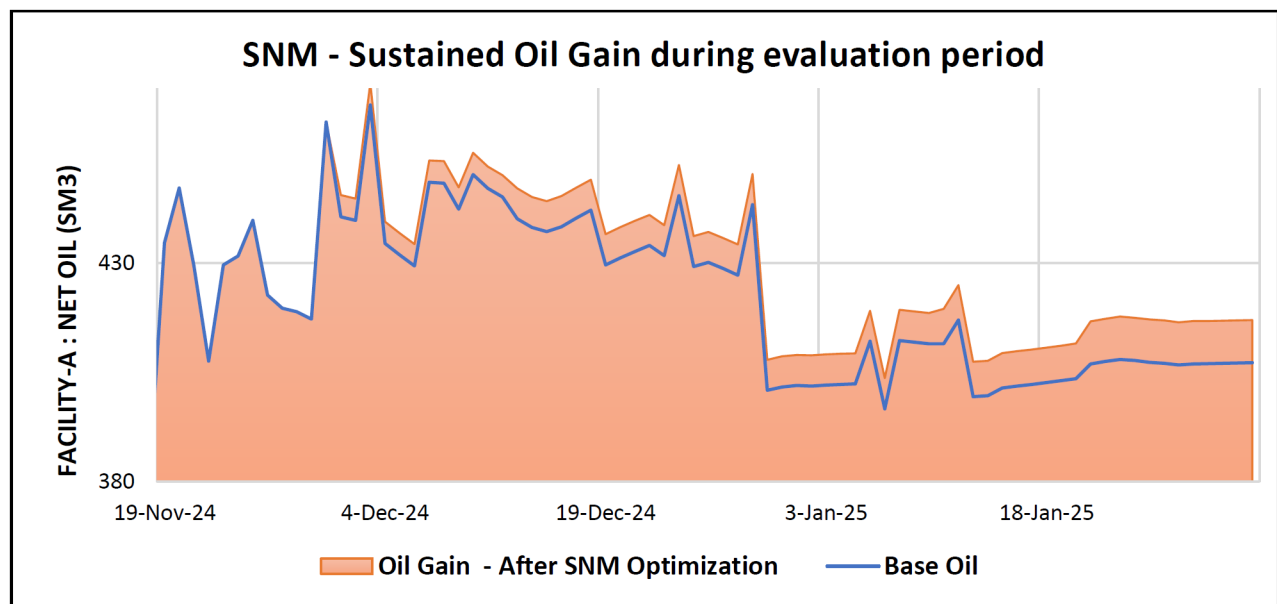


Figure 9—SNM Oil Optimization Results

Other

Operational Efficiency Gains.

- Pipeline Flow Optimization: By identifying underperforming sections of the pipeline, the SNM system allowed to recalibrate these sections, improving flow rates.
- Equipment Utilization: The SNM system identified instances where certain equipment's were either overused or underutilized. By optimizing usage patterns, the system helped balance the workload across the network, ensuring more efficient operations.

Improved Decision-Making.

- Faster Response: Real-time data and alerts basis SNM results allowed quicker issue resolution.

- Better Resource Allocation: Enhanced visibility into equipment performance for more strategic decision-making.
- Scalability: The trial's success enables the SNM to expand to more complex networks, optimizing additional infrastructure.
- Risk Management: Real-time monitoring and predictive maintenance mitigate risks, enhancing network resilience.

Sustained Efficiency and Cost Savings.

- During the evaluation period, real-time monitoring optimized operations, and predictive analysis identified cost-saving opportunities, with the possibility of savings compounding over time.

Time Savings.

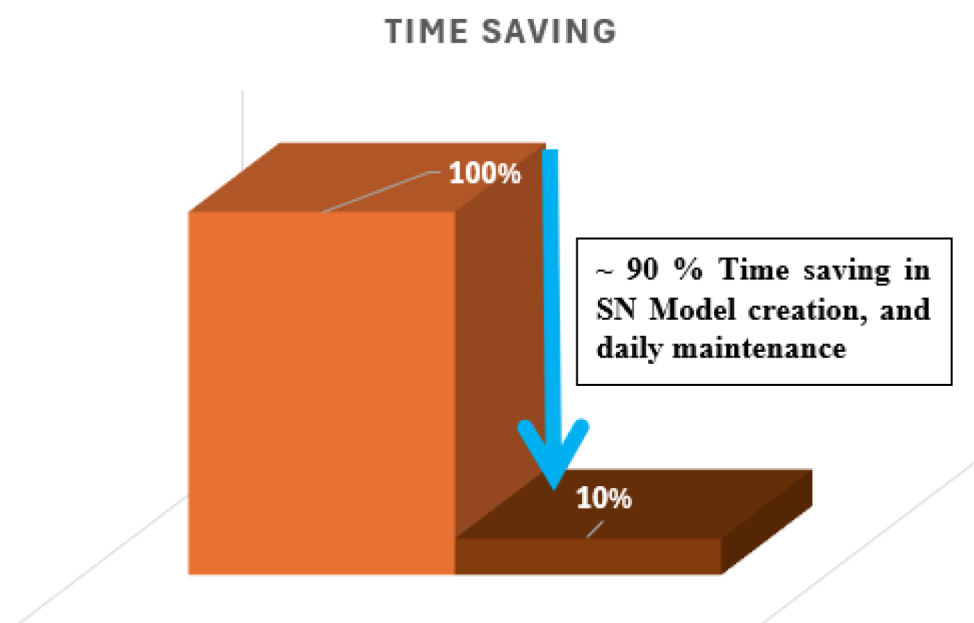


Figure 10—Online SNM – Time Saving Results

Way Forward

Post this trial, the plan would be to roll out the learnings from the Facility-A trial to a pilot project for asset-level deployment across PDO. The Project team will then focus on integrating Machine Learning (ML) and Artificial Intelligence (AI) into the Online Surface Network Model (SNM) to enhance system performance and decision-making.

As automation has already been introduced, ML/AI will be employed to:

- Detect trends and anomalies in real-time, filtering out noise and identifying data quality issues.
- Predict system failure through historical and real-time data.
- Recommend proactive maintenance to optimize performance and reduce downtime.
- Enable closed-loop optimization in real time.
- Improve SNM model convergence algorithms to address issues attributed to mathematical constraint in calculation processes.

The future step will be shifting towards autonomous operations from the results of ML/AI-driven analysis, which will further aid PDO's agenda of enhancing operational efficiency and enabling intelligent, data-driven decision-making.

Change Management

A major challenge in achieving asset-level acceptance of new technology and automation is the way engineers look at new systems and processes. To address this, the project team introduced the Online Surface Network Model (SNM), aiming to shift the work process and promote collaborative working methods. Training played a key role, helping engineers understand both the technical aspects and the broader benefits of the new system. Hands-on workshops, peer mentoring, and feedback loops ensured that engineers felt confident and engaged in the transition.

The online nature of SNM accelerated the change management process, as engineers could collaborate in real-time and integrate the new system into their workflows more quickly than with traditional models. This seamless communication helped overcome adaptation challenges and fostered greater acceptance of the new technology. In addition to technical support, continuous engagement and fostering a sense of ownership were crucial in ensuring the system's smooth implementation and widespread acceptance.

Conclusions

The establishment of the Online Surface Network Model (SNM) on a thin client application with multiple users accessing it at once has led to a 2.97% oil gain, referring to the immediate impact of this technology. This SN model is not a standalone, single-user system but a collaborative platform accessible by multiple users simultaneously, enhancing real-time data sharing and decision-making across teams.

Even though Machine Learning (ML) and Artificial Intelligence (AI) are not yet installed for the SNM technology, their integration in the future will enhance SNM's performance, enhance convergence algorithms, and enable autonomous operations. This will optimize workflows, advance data-driven decision-making, and enhance efficiency further in PDO's E&P operations.

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Nomenclature

AI	= artificial intelligence
BP	= beam pump
DOF	= digital oil field
E&P	= exploration and production
ESP	= electrical submersible pump
ML	= machine learning
PCP	= progressive cavity pump
PDO	= Petroleum Development Oman
QC	= quality check
SN	= surface network
SNM	= surface network model
WMVS	= well management visualization system

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